BUILDING AGENTIC AI APPLICATION CHALLENGE · NOVEMBER 2026

Agentic AI Technical Brief

What the CatchUp AI agent decides, what it can touch, and how we know

The short version

The model decides what to look at. Code decides anything that has to be correct. The student decides anything that changes their week.

What the model decides

- Which of seven tools to call, in what order, with what arguments (up to 8 turns, 90 seconds).
- What to search for in the student's notes, and which retrieved pages to cite.
- Which study items to propose in a draft plan, and which route to suggest.
- When to ask a clarifying question instead of answering.

The seven tools

Tool	What it gives the agent	Can it change data?
get_study_preferences	Time zone, free minutes per weekday, start time	No
list_assignments	Deadlines, estimates, status (titles flagged if they contain instructions)	No
search_course_material	Page-cited passages from the student's own PDFs, or NO_EVIDENCE	No
get_learning_progress	Blocks completed and remaining in the active plan	No
get_student_memory	Notes the student confirmed for the copilot	No
suggest_memory	Proposes one note; unused until the student keeps it	Suggestion only
draft_recovery_plan	Asks the deterministic planner for a route plus up to 6 cited study items	Creates a PENDING draft only

How retrieval stays grounded

- Search runs only over that student's course, under database row-level security.
- Abstains (NO_EVIDENCE) when nothing passes the evidence threshold.
- Passages that try to instruct the AI are flagged and can't back a fact or a study item.
- Each fact is checked against its cited page before display; unsupported facts are removed.

How the plan is calculated

The agent never computes time. draft_recovery_plan hands the work to a deterministic planner that schedules into real free time and reports any shortfall in minutes. The result is stored as a PENDING draft that expires after six hours.

How the student approves

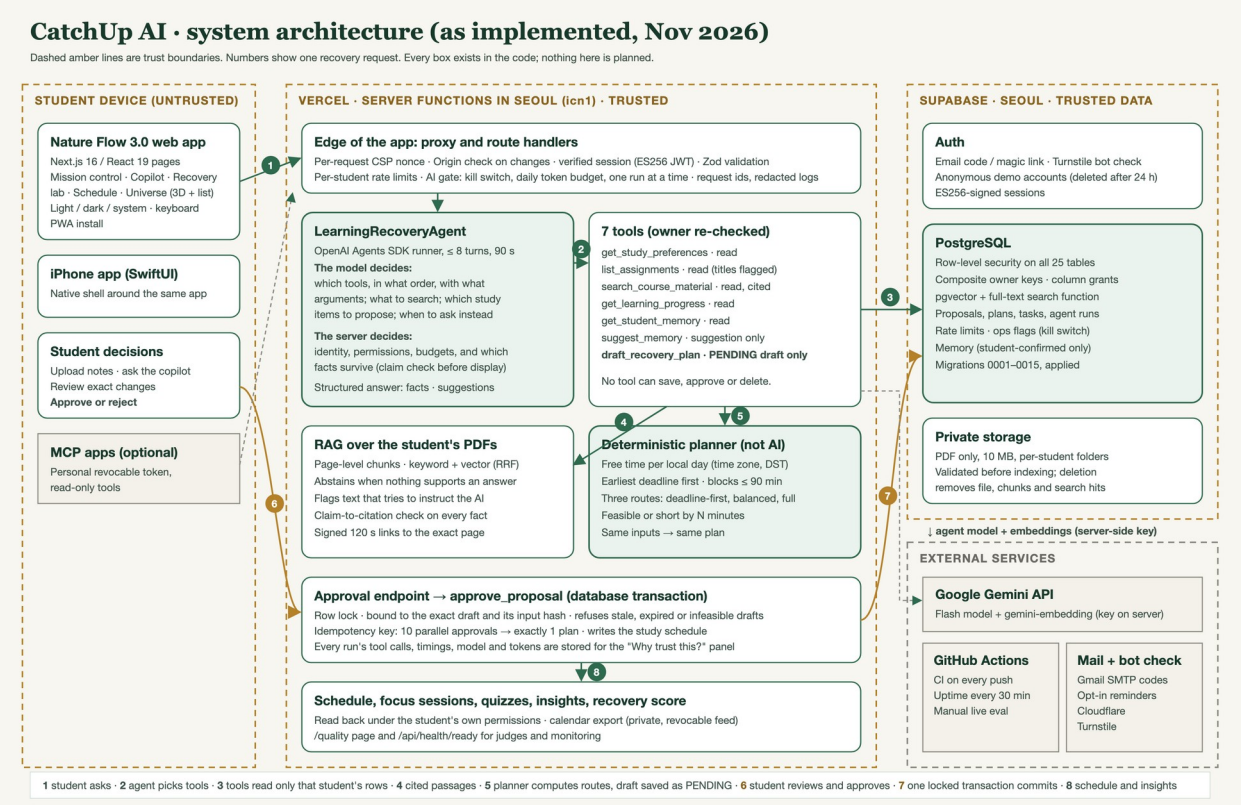
The approval screen lists every block to be added, kept or cancelled. Approval is a locked, idempotent database transaction bound to that exact draft: stale, expired or infeasible drafts are refused, and ten parallel approvals produce exactly one plan.

Errors and unsafe operations

Situation	What happens	Evidence
Invalid tool arguments	Rejected by the schema; the tool never runs	eval B1
Malformed model output	Run marked failed; any draft expired	eval B2
Second draft in one run	Denied	eval B3
Malicious document	Passage flagged; injected 'facts' removed; no draft	eval B4, unit tests
Another student's course id	Tool denied; nothing leaked	eval X2
Model provider failure	Clear error in 0.45 s; nothing saved	live scenario S4
Cost spike or abuse	Kill switch, daily token budget, one run at a time, per-student quotas	integration tests

Evidence

Evidence	Result (4 Nov 2026)	Source
Automated tests	110 unit · 99 integration · 65 end-to-end (production build), all passing; CI on every push	BUILD_STATUS.md, GitHub Actions
Agent evaluation (38 cases)	With the live model (gemini-3.6-flash): 33 passed · 1 failed (Google timed out) · 4 not yet run (daily free quota)	evals/REPORT.md
RAG evaluation (18 questions)	With live Gemini embeddings: Recall@3 100% · MRR 0.939 · multi-page 3/3 · correct abstention 4/4 · 0 cross-user passages · deleted documents return nothing	RAG_QUALITY_REPORT.md
Claim-to-citation check	Precision 0.938, recall 1.0 on 24 labelled claims (one known miss: an antonym)	evals/rag/claims.json
Security	No open Critical or High findings; one real prompt-injection weakness found by the eval and fixed	SECURITY_AUDIT.md, AI_SECURITY_RED_TEAM_REPORT.md
Live performance	Dashboard full load 0.63 s (was 5.0 s); server responses 0.3–0.6 s per page	BUILD_STATUS.md
Live agent traces	Missed lecture: 5 model-chosen tools → feasible draft left pending in 10.7 s, facts cited to pp. 2–3; malicious document resisted; provider failure handled safely	docs/evidence/agent-traces



Architecture with trust boundaries.

What is not proven yet

Four live tool-selection cases and one scenario are still to run on the next day's free quota, and one case (T8) timed out at Google rather than answering. Answers take 10–90 s on the shared free tier. No live answers have been reviewed by an independent person yet.